

## STORHOFDI, Iceland

WMO: 040480

Lat: 63.400N

Lon: 20.288W

Elev: 387

StdP: 14.49

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 2             | 20.4       | 22.9 | 8.4                           | 8.6 | 21.9 | 11.4 | 10.0 | 24.3 | 76.9                  | 33.4 | 69.1 | 35.0 | 25.6                     | 0    | 1.126 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 5.3                    | 57.5            | 53.0 | 55.8 | 52.3 | 54.5 | 51.7 | 54.5                | 56.1 | 53.4 | 54.7 | 52.5 | 53.8 | 12.6                 | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 53.6                            | 62.0 | 54.8 | 52.6 | 59.7 | 53.9 | 51.8 | 57.9 | 53.1 | 23.0          | 56.2 | 22.4 | 54.9 | 21.9 | 53.9 | 61.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 64.7          | 57.4 | 51.0 | 16.1                       | 62.1 | 3.5                | 3.2 | 13.6                                               | 64.4 | 11.5       | 66.3 | 9.5        | 68.1 | 7.0        | 70.5 |    |    |
| (5)               |      |      | 13.4                       | 56.3 | 3.4                | 2.0 | 11.0                                               | 57.7 | 9.0        | 58.9 | 7.1        | 60.0 | 4.7        | 61.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 42.1   | 36.2  | 35.5  | 35.7  | 39.2  | 43.5  | 48.0  | 51.4  | 51.3  | 47.4  | 41.6  | 38.2  | 36.3 |  |  |
|                                                                        | DBStd       | 7.35   | 5.34  | 5.67  | 5.57  | 4.57  | 3.21  | 2.57  | 2.20  | 2.17  | 3.61  | 4.78  | 5.24  | 5.74 |  |  |
|                                                                        | HDD50       | 3019   | 428   | 407   | 443   | 324   | 201   | 68    | 10    | 10    | 89    | 260   | 354   | 425  |  |  |
|                                                                        | HDD65       | 8373   | 893   | 827   | 908   | 774   | 665   | 510   | 423   | 425   | 529   | 725   | 804   | 890  |  |  |
|                                                                        | CDD50       | 120    | 0     | 0     | 0     | 0     | 0     | 8     | 52    | 50    | 10    | 0     | 0     | 0    |  |  |
|                                                                        | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |  |  |
|                                                                        | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |  |  |
|                                                                        | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |  |  |
|                                                                        | WSAvg       | 23.1   | 27.0  | 27.8  | 26.6  | 24.9  | 19.8  | 17.7  | 16.5  | 17.7  | 23.2  | 23.3  | 26.4  | 26.2 |  |  |
|                                                                        | PrecAvg     | 61.9   | 5.9   | 5.4   | 5.5   | 4.3   | 3.9   | 3.9   | 3.8   | 5.3   | 5.5   | 6.1   | 5.7   | 5.8  |  |  |
| (15) Precipitation                                                     | PrecMax     | 80.1   | 10.6  | 11.9  | 11.9  | 8.5   | 9.7   | 7.6   | 8.1   | 10.6  | 11.5  | 11.1  | 13.0  | 9.3  |  |  |
|                                                                        | PrecMin     | 47.2   | 1.8   | 1.0   | 0.4   | 1.5   | 0.4   | 0.5   | 1.5   | 0.2   | 1.7   | 2.0   | 1.1   | 2.2  |  |  |
|                                                                        | PrecStd     | 7.9    | 2.2   | 2.3   | 2.3   | 1.5   | 2.0   | 1.9   | 1.5   | 2.6   | 2.2   | 2.1   | 2.5   | 1.9  |  |  |
|                                                                        | PrecStd     | 7.9    | 2.2   | 2.3   | 2.3   | 1.5   | 2.0   | 1.9   | 1.5   | 2.6   | 2.2   | 2.1   | 2.5   | 1.9  |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4% DB     | 46.5   | 46.8  | 46.5  | 48.3  | 53.0  | 57.4  | 60.0  | 61.5  | 54.9  | 51.2  | 48.6  | 47.8  |      |  |  |
|                                                                        | 0.4% MCWB   | 45.6   | 46.2  | 45.5  | 45.0  | 48.1  | 50.7  | 54.0  | 55.0  | 52.5  | 49.1  | 48.1  | 46.9  |      |  |  |
|                                                                        | 2% DB       | 45.5   | 45.8  | 45.5  | 46.6  | 50.4  | 54.7  | 57.8  | 57.3  | 53.7  | 50.1  | 47.5  | 46.6  |      |  |  |
|                                                                        | 2% MCWB     | 44.5   | 45.1  | 44.5  | 44.6  | 46.2  | 50.1  | 53.3  | 53.2  | 51.8  | 48.6  | 46.8  | 45.8  |      |  |  |
|                                                                        | 5% DB       | 44.6   | 44.7  | 44.6  | 45.7  | 48.9  | 52.9  | 56.3  | 55.6  | 52.7  | 49.1  | 46.4  | 45.4  |      |  |  |
|                                                                        | 5% MCWB     | 43.5   | 43.8  | 43.6  | 43.9  | 45.5  | 49.3  | 52.6  | 52.5  | 51.0  | 47.8  | 45.4  | 44.4  |      |  |  |
|                                                                        | 10% DB      | 43.3   | 43.2  | 43.3  | 44.9  | 47.7  | 51.6  | 54.8  | 54.3  | 51.7  | 48.0  | 45.2  | 44.1  |      |  |  |
|                                                                        | 10% MCWB    | 42.0   | 41.9  | 41.9  | 43.2  | 45.0  | 48.7  | 51.9  | 51.8  | 50.2  | 46.7  | 43.9  | 42.7  |      |  |  |
|                                                                        | 0.4% WB     | 46.2   | 46.6  | 45.9  | 46.7  | 49.3  | 52.6  | 55.8  | 56.4  | 54.0  | 50.4  | 48.5  | 47.6  |      |  |  |
|                                                                        | 0.4% MCDB   | 46.3   | 46.6  | 46.0  | 47.0  | 51.8  | 55.0  | 58.4  | 59.1  | 54.2  | 50.6  | 48.6  | 47.7  |      |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2% WB       | 45.0   | 45.4  | 45.0  | 45.7  | 47.6  | 51.2  | 54.4  | 54.8  | 52.7  | 49.2  | 47.2  | 46.2  |      |  |  |
|                                                                        | 2% MCDB     | 45.4   | 45.6  | 45.2  | 46.1  | 49.0  | 53.1  | 56.4  | 56.3  | 53.1  | 49.6  | 47.3  | 46.5  |      |  |  |
|                                                                        | 5% WB       | 43.8   | 44.0  | 43.9  | 44.8  | 46.7  | 50.3  | 53.4  | 53.6  | 51.5  | 48.2  | 45.7  | 44.7  |      |  |  |
|                                                                        | 5% MCDB     | 44.4   | 44.5  | 44.4  | 45.3  | 47.9  | 52.0  | 55.0  | 54.7  | 52.1  | 48.9  | 46.1  | 45.3  |      |  |  |
|                                                                        | 10% WB      | 42.1   | 42.2  | 42.2  | 43.7  | 46.0  | 49.3  | 52.6  | 52.6  | 50.6  | 46.9  | 44.2  | 43.0  |      |  |  |
|                                                                        | 10% MCDB    | 43.1   | 43.1  | 43.2  | 44.6  | 47.2  | 50.9  | 54.2  | 53.7  | 51.5  | 47.8  | 45.1  | 43.9  |      |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        | 5.7    | 6.0   | 6.0   | 5.9   | 6.0   | 5.5   | 5.3   | 4.9   | 4.7   | 5.0   | 5.3   | 5.7   |      |  |  |
|                                                                        | 5% DB MCDBR | 5.4    | 6.0   | 5.5   | 6.0   | 7.3   | 7.8   | 8.0   | 7.1   | 4.9   | 4.6   | 5.2   | 5.3   |      |  |  |
|                                                                        | 5% DB MCWBR | 6.5    | 7.3   | 6.5   | 5.7   | 5.2   | 5.5   | 5.7   | 5.2   | 4.8   | 5.3   | 6.3   | 6.7   |      |  |  |
|                                                                        | 5% WB MCDBR | 5.8    | 5.8   | 5.5   | 4.8   | 5.6   | 6.4   | 6.4   | 5.3   | 4.0   | 4.4   | 5.4   | 5.5   |      |  |  |
|                                                                        | 5% WB MCWBR | 7.0    | 7.2   | 6.8   | 5.6   | 4.6   | 5.0   | 5.1   | 4.5   | 4.7   | 5.4   | 6.6   | 6.9   |      |  |  |
|                                                                        | 5% WB MCWBR | 7.0    | 7.2   | 6.8   | 5.6   | 4.6   | 5.0   | 5.1   | 4.5   | 4.7   | 5.4   | 6.6   | 6.9   |      |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.410  | 0.312 | 0.331 | 0.361 | 0.359 | 0.352 | 0.354 | 0.341 | 0.343 | 0.306 | 0.374 | 0.373 |      |  |  |
|                                                                        | taud        | 2.659  | 2.233 | 2.340 | 2.300 | 2.383 | 2.461 | 2.485 | 2.529 | 2.446 | 2.332 | 2.772 | 2.592 |      |  |  |
|                                                                        | Ebn at Noon | 105    | 193   | 242   | 259   | 271   | 276   | 272   | 266   | 238   | 195   | 120   | 60    |      |  |  |
|                                                                        | Edh at Noon | 10     | 19    | 25    | 33    | 33    | 31    | 30    | 26    | 23    | 17    | 9     | 6     |      |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 43     | 181   | 499   | 1029  | 1503  | 1587  | 1457  | 1164  | 676   | 290   | 71    | 22    |      |  |  |
|                                                                        | RadStd      | 5      | 30    | 70    | 124   | 150   | 195   | 121   | 84    | 68    | 29    | 11    | 1     |      |  |  |

## Historical Trends

|      |                        | Heating |        | Cooling |       |       | Degree-Days |       |       |       |     |      |
|------|------------------------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|-----|------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |     |      |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m) |      |
| (46) | Station Only           | +1.17   | +2.02  | N/A     | +2.39 | +1.24 | +1.08       | -373  | -468  | +99   | N/A | (46) |
| (47) | Regional (5 neighbors) | +0.92   | +1.87  | +2.38   | +1.15 | N/A   | N/A         | -288  | -347  | +67   | N/A | (47) |

Nomenclature: See separate page